

Excel Project Handout

Project Title: Health Records – From Raw Data to Insights

Background

You have been provided with a dataset of health records containing patient visits, diagnoses, doctors, and treatment costs. Unfortunately, the data is messy and not ready for analysis. Your job is to clean, format, analyse, and report using Excel skills you've Learned.

Objectives

- Clean and organise messy data
- Apply Excel formatting for readability
- Use formulas across different categories (Logical, Statistical, Lookup, Math & Date)
- Build PivotTables and PivotCharts for analysis
- Present your findings clearly

Part 1: Data Cleaning

- Remove any duplicate records based on PatientID.
- Convert VisitDate into a proper Excel date format (dd-mmm-yyyy).
- Handle missing data: Replace blank Diagnosis with 'Unknown'; Replace blank Doctor with 'Unassigned'.
- Remove the empty column AGE RANGE (you will rebuild this later).
- Standardise text values: Ensure Gender only has Male, Female, Other; Remove extra spaces from Doctor, Treatment, and Diagnosis fields.

Part 2: Formatting

- Apply bold text + background colour to column headers.
- Format Cost as currency with 2 decimal places.
- Format VisitDate as dd-mmm-yyyy.
- Add conditional formatting: Highlight patients older than 60; Highlight treatments costing more than 3000

Part 3: Formulas

- Create a new sheet called Formulas and answer the following:
- Logical Functions: Use IF to create a new column 'Age Group'.
- Statistical Functions: Find average, max, min of Cost; Count patients with Diabetes.
- Lookup & Reference: Create lookup table with Treatment Types and Standard Costs; use VLOOKUP.
- Math & Trig: Add a column 'Cost with Tax' = Cost $\times$ 1.1; Use ROUND, ROUNDUP, ROUNDDOWN.

- Date Functions: Extract Year and Month from VisitDate; Rebuild AGE RANGE with IF

Part 4: Pivot Tables & Charts

- Create a new sheet called Analysis.
- PivotTable: Total Cost by Doctor.
- PivotTable: Count of Patients by Gender.
- PivotTable: Average Cost by Diagnosis.
- PivotTable: Age Group Distribution.
- Insert PivotCharts: Pie chart (Gender distribution), Column chart (Total Cost by Diagnosis)

Deliverables

- Raw Data (original sheet)
- Cleaned Data sheet
- Formulas sheet (all your calculations)
- Analysis sheet (PivotTables & Charts)

Grading Rubric (100%)

- Data Cleaning – 20%
- Formatting – 15%
- Formulas – 30%
- PivotTables & Charts – 25%
- Presentation (clarity, neatness) – 10%

Dataset: [Health Dirty Dataset](#)